

EXTREMAL GRAPH THEORY

Erdős Problem 915

Maximizing Edge Density Subject to Connectivity Constraints

One-hour version

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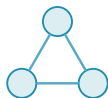
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KU Leuven, Master of Mathematics

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The eight models



simple graph

one edge per pair



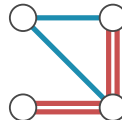
multigraph

parallel edges $\mu(u, v) \geq 1$



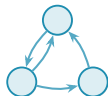
hypergraph

hyperedges span $r \geq 3$



multihypergraph

a hyperedge repeated



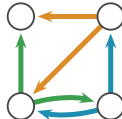
simple digraph

one arc per ordered pair



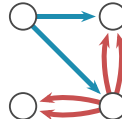
directed multigraph

parallel and opposite arcs



directed hypergraph

two share an edge

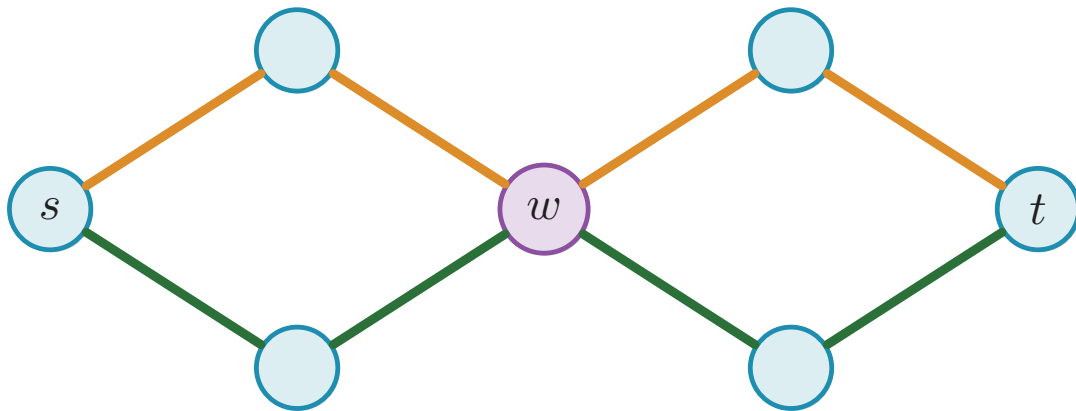


directed multihypergraph

a hyperedge repeated

The eight models side by side, undirected on top and directed below. The simple graph is a triangle, the multigraph has parallel edges, the hypergraph has three hyperedges, and the multihypergraph has one repeated hyperedge. The directed versions have arcs instead of edges, with the same pattern of repetition and hyperedges.

The separation axis

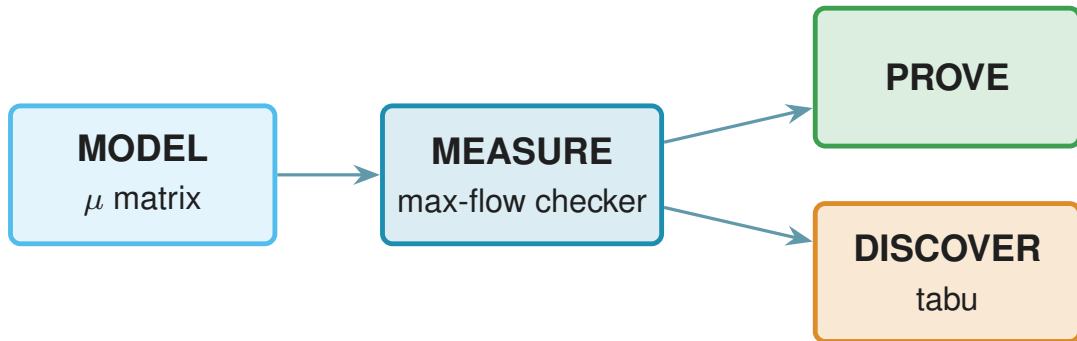


Two edge-disjoint s – t routes both pass through w , so $\lambda(s, t) = 2$ but $\kappa(s, t) = 1$.

Notation for the extremal functions

Model	edge-disjoint routes	internally vertex-disjoint routes
Simple graph	$\ell_m(n)$	$k_m(n)$
Multigraph	$L_m(n)$	$K_m(n)$
r -uniform simple hypergraph	$\ell_m^{(r)}(n)$	$k_m^{(r)}(n)$
r -uniform multihypergraph	$L_m^{(r)}(n)$	$K_m^{(r)}(n)$
Simple digraph	$\ell_m^{\text{dir}}(n)$	$k_m^{\text{dir}}(n)$
Directed multigraph	$L_m^{\text{dir}}(n)$	$K_m^{\text{dir}}(n)$
Directed r -uniform simple hypergraph	$\ell_m^{\text{dir}(r)}(n)$	$k_m^{\text{dir}(r)}(n)$
Directed r -uniform multihypergraph	$L_m^{\text{dir}(r)}(n)$	$K_m^{\text{dir}(r)}(n)$

The architecture

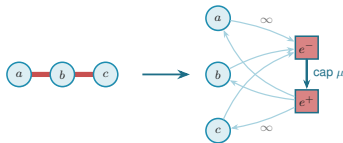


A single model feeds the exact max-flow checker. Certification turns that measurement into finite upper bounds, while heuristic search uses it to discover lower bounds.

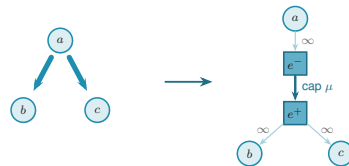
The building blocks of the connectivity checker



Split each vertex v into v^{in} and v^{out} joined by a unit-capacity arc to transform vertex-disjointness into edge-disjointness.

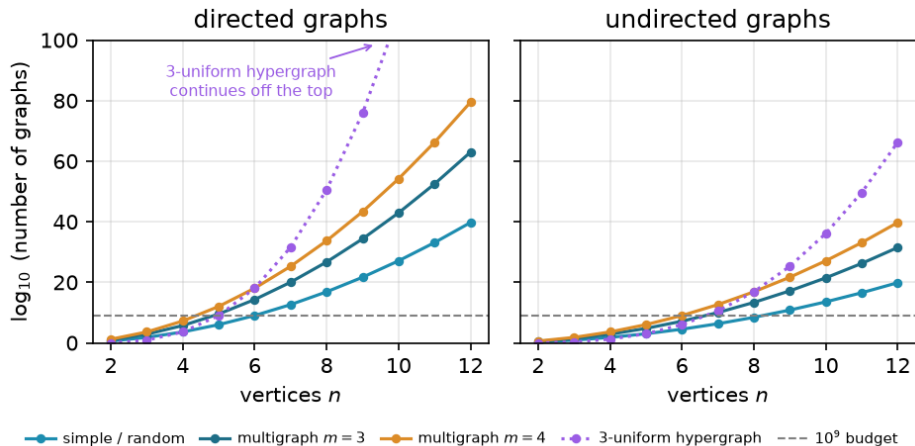


Transform a hyperedge into a gate with capacity equal to the multiplicity of the hyperedge.



The directed hyperedge is transformed into a gate with one tail and two heads. This gate has capacity equal to the multiplicity of the hyperedge and every route must enter at the tail and leave at a head.

Candidate counts

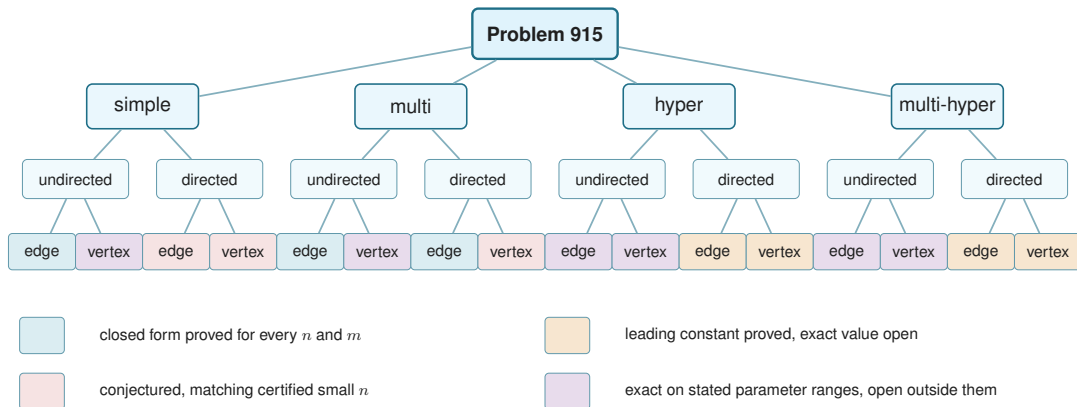


Candidate counts for blind enumeration on a logarithmic scale. Every model crosses the generous 10^9 budget in single-digit order, and directed objects grow fastest because they have more independent entries to decide.

Generating search: a benchmark

n	$\ell_2^{\text{dir}}(n)$	blind (s)	pruned (s)	generated (s)
3	4	0.05	0.003	0.009
4	6	6.3	0.008	0.008
5	8	—	0.22	0.039
6	10	—	14.5	0.73
7	12	—	1156.6	17.0

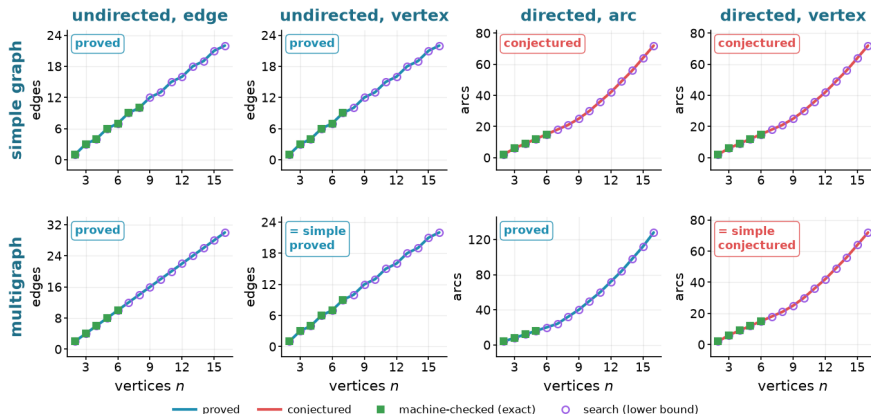
Status of the sixteen variants



Status of the sixteen variants. Colour marks the status of a leaf: blue a closed form proved for every n and m , purple an exact value proved on stated parameter ranges and open outside them, red a conjectured exact formula with a proved lower bound, and amber a proved leading term with the exact value unknown.

Eight graph variants at $m = 3$

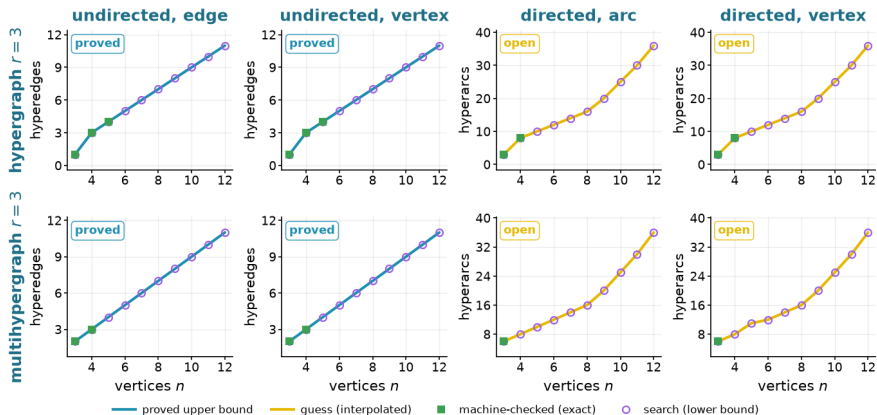
Erdős 915 across the sixteen variants, $m = 3$ · simple and multigraph models



The eight graph-model variants at $m = 3$: simple and multigraph rows, undirected and directed columns. A blue curve is a theorem, a red one a conjecture and a yellow one an interpolation, filled squares are exact finite values, and open circles are witnesses.

Eight hypergraph variants at $m = 3$

Erdős 915 across the sixteen variants, $m = 3$ · hypergraph and multihypergraph models

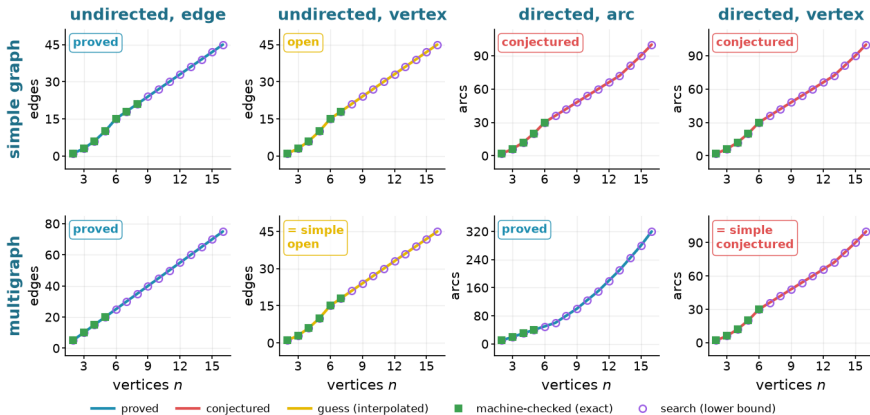


The eight hypergraph-model variants at $m = 3$: hypergraph and multihypergraph rows, undirected and directed columns, same colour and marker key.

The blue curve is a proved upper bound rather than a proved value, attained only inside the stated ranges, so a square or circle below it marks a gap that is open rather than an error.

Eight graph variants at $m = 6$

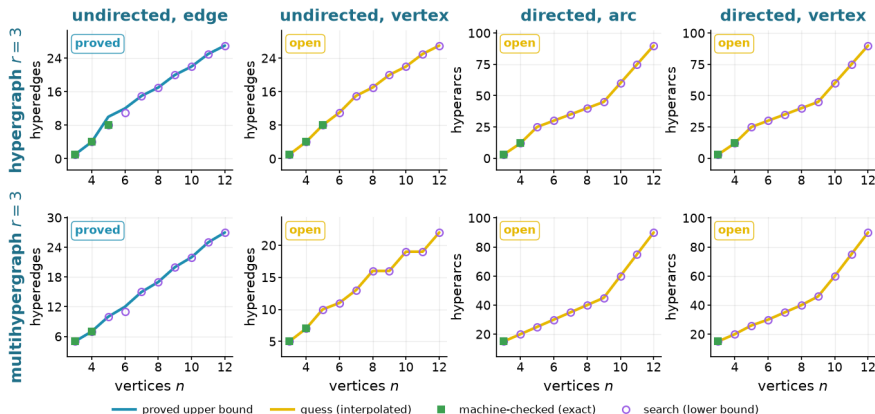
Erdős 915 across the sixteen variants, $m = 6$ · simple and multigraph models



The same eight graph-model variants at $m = 6$. Raising m widens every feasible band, and the exact squares thin out as the enumeration range falls behind the constructions.

Eight hypergraph variants at $m = 6$

Erdős 915 across the sixteen variants, $m = 6$ · hypergraph and multihypergraph models



The same eight hypergraph-model variants at $m = 6$. The blue curve is again the proved upper bound, and at this m it runs visibly above what has been reached: the exact square at $n = 5$ is 8 against the curve's 10, and at $n = 6$ the curve reads 12 where the true maximum is 11.

The sixteen variants, summarised

Model	Edge or arc-disjoint	Vertex-disjoint
<i>Undirected</i>		
simple graph	$\lfloor m(n-1)/2 \rfloor$, proved	the edge value for $m \leq 4$, and $\lfloor 8n/3 \rfloor - 4$ at $m = 5$; open for $m \geq 6$
multigraph	$(m-1)(n-1)$, proved	the simple value
r -hypergraph	$\leq \lfloor (m-1)(n-1)/(r-1) \rfloor$, with equality in the range below	$\lfloor (n-1)/(r-1) \rfloor$ at $m = 2$ and $\leq \lfloor 2(n-1)/(r-1) \rfloor$ at $m = 3$; open for $m \geq 4$
r -multi-hyper	the same bound, with equality when $(r-1) \mid (n-1)$	the same bounds, repeats closing cases the simple model misses at $m = 3$
<i>Directed</i>		
simple digraph	$M(n)$ at $m = 2$, proved; $n^2/4 + \Theta_m(n)$ for $m \geq 3$, exact value conjectured	$M(n)$ at $m = 2$, proved; $n^2/4 + \Theta_m(n)$ for $m \geq 3$, equality with the arc value open
multigraph	$(m-1)M(n)$, proved	the simple digraph value
r -hypergraph	$(1+o(1))(m-1)n^2/[4(r-1)]$ for fixed r, m , leading term proved	the same leading term for fixed r, m
r -multi-hyper	the same leading term, the bound proved with repeats allowed	the same leading term for fixed r, m

Exact values, $m = 3$

Model	Variant	$n = 2$	$n = 3$	$n = 4$	$n = 5$	$n = 6$	$n = 7$	$n = 8$
simple graph	undirected, edge ($\ell_m(n)$), proved	1	3	4	6	7	9	10
	undirected, vertex ($k_m(n)$), proved	1	3	4	6	7	9	10
	directed, arc ($\ell_m^{\text{dir}}(n)$), conjectured	2	6	9	12	15	≥ 18	≥ 21
	directed, vertex ($k_m^{\text{dir}}(n)$), conjectured	2	6	9	12	15	≥ 18	≥ 21
multigraph	undirected, edge ($L_m(n)$), proved	2	4	6	8	10	12	14
	undirected, vertex ($k_m(n)$), = simple, proved	1	3	4	6	7	9	10
	directed, arc ($L_m^{\text{dir}}(n)$), proved	4	8	12	16	20	24	32
	directed, vertex ($k_m^{\text{dir}}(n)$), = simple, conjectured	2	6	9	12	15	≥ 18	≥ 21
hypergraph $r = 3$	undirected, edge, proved		1	3	4	5	6	7
	undirected, vertex, proved		1	3	4	5	6	7
	directed, arc, open		3	8	≥ 10	≥ 12	≥ 14	≥ 16
	directed, vertex, open		3	8	≥ 10	≥ 12	≥ 14	≥ 16
multihypergraph $r = 3$	undirected, edge, proved		2	3	4	5	6	7
	undirected, vertex, proved		2	3	4	5	6	7
	directed, arc, open		6	≥ 8	≥ 10	≥ 12	≥ 14	≥ 16
	directed, vertex, open		6	≥ 8	≥ 10	≥ 12	≥ 14	≥ 16

Exact values, $m = 6$

Model	Variant	$n = 2$	$n = 3$	$n = 4$	$n = 5$	$n = 6$	$n = 7$	$n = 8$
simple graph	undirected, edge ($\ell_m(n)$), proved	1	3	6	10	15	18	21
	undirected, vertex ($k_m(n)$), open	1	3	6	10	15	18	≥ 21
	directed, arc ($\ell_m^{\text{dir}}(n)$), conjectured	2	6	12	20	30	≥ 36	≥ 42
	directed, vertex ($k_m^{\text{dir}}(n)$), conjectured	2	6	12	20	30	≥ 36	≥ 42
multigraph	undirected, edge ($L_m(n)$), proved	5	10	15	20	25	30	35
	undirected, vertex ($k_m(n)$), = simple, open	1	3	6	10	15	18	≥ 21
	directed, arc ($L_m^{\text{dir}}(n)$), proved	10	20	30	40	50	60	80
	directed, vertex ($k_m^{\text{dir}}(n)$), = simple, conjectured	2	6	12	20	30	≥ 36	≥ 42
hypergraph $r = 3$	undirected, edge, proved		1	4	8	≤ 12	15	17
	undirected, vertex, open		1	4	8	≥ 11	≥ 15	≥ 17
	directed, arc, open		3	12	≥ 25	≥ 30	≥ 35	≥ 40
	directed, vertex, open		3	12	≥ 25	≥ 30	≥ 35	≥ 40
multihypergraph $r = 3$	undirected, edge, proved		5	7	10	≤ 12	15	17
	undirected, vertex, open		5	7	≥ 10	≥ 11	≥ 13	≥ 16
	directed, arc, open		15	≥ 20	≥ 25	≥ 30	≥ 35	≥ 40
	directed, vertex, open		15	≥ 20	≥ 26	≥ 30	≥ 35	≥ 40

What remains open

Problem	Known here	Remaining target
Directed simple arc, $m \geq 3$	$\ell_m^{\text{dir}}(n) = n^2/4 + \Theta_m(n)$	Prove the decomposition, or otherwise determine the linear coefficient.
Directed simple vertex, $m \geq 3$	$k_m^{\text{dir}}(n) = n^2/4 + \Theta_m(n)$	Decide whether its exact value equals the arc value, given that an arc upper bound does not transfer.
Directed multigraph classification	Exact value for all n, m and quadratic-branch uniqueness for $n \geq \max(8, 2m)$	Classify the range $n < 2m$ and the remaining linear-branch extremisers.
Undirected simple vertex, $m \geq 6$	Block formulation and $m/2 < c_m \leq 2(m-1)$, where $c_m = \lim_n k_m(n)/(n-1)$	Determine c_m , whose likely next structure is triconnected rather than merely blockwise.
Multigraph vertex, parallel copies distinct	Block-knapsack formulation and order $\Theta(m^2 n)$ as $m \rightarrow \infty$ with $n/m \rightarrow \infty$, with the per-vertex constant between $3 - 2\sqrt{2}$ and 2	Determine the asymptotic constant and the optimal 2-connected blocks.
Hypergraph vertex, $m \geq 4$	Exact at $m = 2$ and, in the stated range, $m = 3$	Locate the first failure for $r \geq 3$ and replace the rank-two decomposition used at $m = 3$.
Directed hypergraphs	For fixed r, m , leading term $(m-1)n^2/(4(r-1))$ for all orientations and both separations	Determine finite exact values and whether forward and general agree once $n > r$.

AI: curse or blessing?

Pros

- $\text{\LaTeX}$ typesetting
- length & complexity of generated text

Cons

- length & complexity of generated text
 - control
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It's a tool.

short leash + AI proofs = argued conjecture [AI]